← Back **Graded Quiz**
Graded Assignment • 30 min

Due Mar 16, 11:59 PM CET

1. The amount of information obtained by knowing the result of a random experiment is the same to the amount of (A)_____ on the result. In the case of tossing coin it is maximized when the coin is (B)_____ and the corresponding amount of information is (C)___ bit.

1 point

- (A) certainty, (B) fair, (C) 1
- (A) certainty, (B) fair, (C) 0
- (A) certainty, (B) unfair, (C) 1
- (A) uncertainty, (B) fair, (C) 1
- (A) uncertainty, (B) unfair, (C) 0

2.
- Suppose that the maximum frequency component of a signal is W (Hz). According to the sampling theorem, if the sample rate is above or equal to (A)_____, called the Nyquist rate, the original signal can be recovered perfectly.
- Representing a continuous value requires (B)_____ number of bits because the resolution is (B)_____. If we increase the resolution of a quantizer, the distortion is (C)_____ but the information rate will be (D)_____.

1 point

- (A) W, (B) finite, (C) increased, (D) decreased
- (A) W, (B) infinite, (C) decreased, (D) increased
- (A) $2W$, (B) finite, (C) increased, (D) decreased
- (A) $2W$, (B) infinite, (C) increased, (D) decreased
- (A) $2W$, (B) infinite, (C) decreased, (D) increased

3.
- For wireless communications, we use an electrical device called antenna which converts an electrical current into a(n) (A)_____ and vice versa. A(n) (A)_____ can propagate a long distance with the speed of light. By doing this, we can deliver information wirelessly.
- Mathematically, we can decompose a signal according to its constituent (B)_____ components. The power spectrum of a signal indicates the power distribution of the information bearing signal according to (B)_____.
- Here, the range of non-zero power distribution is called the signal (C)_____ and it determines the speed of change in the signal. Remember that this is one resource we need to pay for transferring information.
- If we increase the (D)_____ of the transmit signal, the intensity of the propagated radio wave increases, which implies longer propagation distance and better immunity for non-ideal effects. So, (D)_____ is another resource we need to pay.

1 point

- (A) radio wave, (B) frequency, (C) power, (D) bandwidth
- (A) radio wave, (B) frequency, (C) bandwidth, (D) power
- (A) radio wave, (B) time, (C) power, (D) bandwidth
- (A) acoustic wave, (B) frequency, (C) bandwidth, (D) power
- (A) acoustic wave, (B) time, (C) power, (D) bandwidth

4. The PSD of the thermal noise is (A)_____. The noise power contained within the bandwidth of W is (B)_____. Using the bandwidth of W is equivalent to use (C)_____ independent samples. Then, the Shannon capacity is given as (D)_____, when the received signal power is P.

1 point

- (A) $\frac{N_0}{2}$, (B) $\frac{N_0}{2}W$, (C) W. (D) $\frac{W}{2}\log_2(1+\frac{2P}{N_0W})$
- (A) $\frac{N_0}{2}$, (B) N_0W, (C) $2W$. (D) $\frac{W}{2}\log_2(1+\frac{P}{N_0W})$
- (A) $\frac{N_0}{2}$, (B) N_0W, (C) $2W$. (D) $W\log_2(1+\frac{P}{N_0W})$
- (A) N_0, (B) N_0W, (C) W. (D) $\frac{W}{2}\log_2(1+\frac{P}{N_0W})$
- (A) N_0, (B) $2N_0W$, (C) $2W$. (D) $W\log_2(1+\frac{P}{N_0W})$

5. What is the right order of processing sequence at the transmitter?

1 point

- (Channel Encoding) – (Constellation Mapping) – (Waveform Mapping) – (Up Converting & Power Amp)
- (Channel Encoding) – (Waveform Mapping) – (Constellation Mapping) – (Up Converting & Power Amp)
- (Waveform Mapping) – (Constellation Mapping) – (Channel Encoding) – (Up Converting & Power Amp)
- (Constellation Mapping) – (Waveform Mapping) – (Up Converting & Power Amp) – (Channel Encoding)
- (Constellation Mapping) – (Channel Encoding) – (Up Converting & Power Amp) – (Waveform Mapping)

6. Consider the case where a detection fails but the channel decoder recovers it. The noisy received symbols close to the (A)_____ are very probable to produce bit errors. But, the channel decoder may find that the erroneous bit sequence is not a possible candidate sequence and correct the errors by mapping to the closest (B)_____ sequence. Although such mapping can fail either, such probability vanishes as the information bit size k (C)_____.

1 point

- (A) original symbols, (B) erroneous, (C) increases
- (A) original symbols, (B) possible, (C) decreases
- (A) original symbols, (B) possible, (C) increases
- (A) boundaries, (B) erroneous, (C) decreases
- (A) boundaries, (B) possible, (C) increases

7. Select the correct answer(s).

1 point

- [ ] If we divide the area into many cells and use one base station per each cell, much more power is required.
- [x] If users in different cells reuse the same frequency channels, the required bandwidth becomes much reduced.
- [ ] Supporting many users in a large area by only one transmitter is the power efficient way.
- [ ] Cellular structure is bandwidth-efficient way to support many users in a large area but is not power-efficient.

8. The base station selects one among the predetermined (A)_____ based on the (B)_____ and transferring information by using it. This is called (C)_____.

1 point

- (A) AMC (B) MCS (C) CQI
- (A) AMC (B) CQI (C) MCS
- (A) MCS (B) AMC (C) CQI
- (A) MCS (B) CQI (C) AMC
- (A) CQI (B) MCS (C) AMC

9.
- In frequency division multiple access (FDMA), (A)_____ channels are allocated by dividing frequency
- In time division multiple access (TDMA), (B)_____ channels are allocated by dividing time.
- In code division multiple access (CDMA), typically (C)_____ codes are allocated to users so that each user can use the whole time and bandwidth.

1 point

- (A) orthogonal (B) orthogonal (C) orthogonal
- (A) orthogonal (B) orthogonal (C) quasi-orthogonal
- (A) orthogonal (B) quasi-orthogonal (C) orthogonal
- (A) quasi-orthogonal (B) orthogonal (C) quasi-orthogonal
- (A) quasi-orthogonal (B) quasi-orthogonal (C) quasi-orthogonal

10. Compare the two-user rate region of CDMA, FDMA, and TDMA.

1 point

- CDMA ≥ FDMA ≥ TDMA
- CDMA ≥ TDMA ≥ FDMA
- FDMA ≥ CDMA ≥ TDMA
- FDMA ≥ TDMA ≥ CDMA
- TDMA ≥ FDMA ≥ CDMA